Start with what we can observe

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Observation / ES: observación / PT: observação

Something we record or measure, like a position at a time.

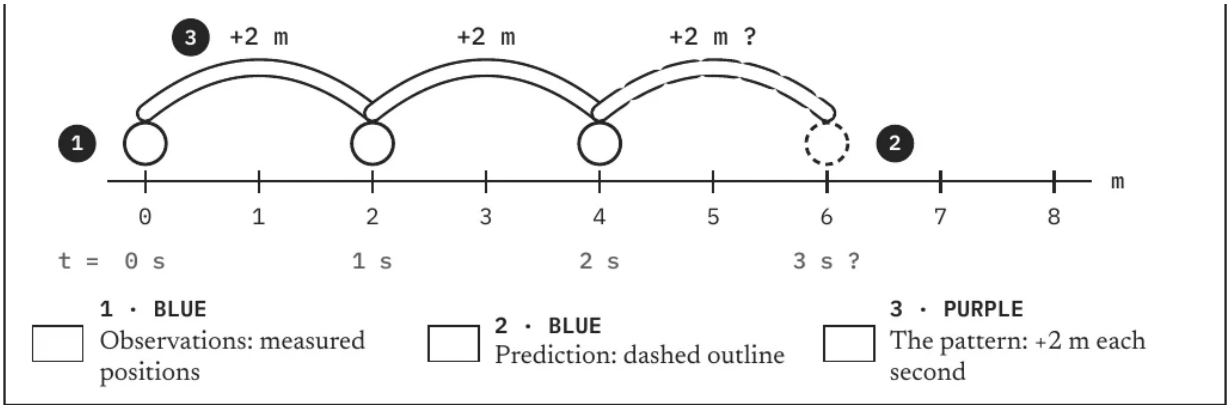
Also on your note page: Model, Prediction, Testable question.

B READ THE RELATIONSHIP

An observation is something you see or measure. A prediction says what you expect later.

Position tells where. Velocity tells how fast and which way.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart is at 2 m, 3 m, then 4 m, one second apart. Predict its position one second later.

Given x = 2, 3, 4 m 1 s between readings

Rule next = last + change each second

Work change = 3 − 2 = +1 m next = 4 m + 1 m

Answer 5 m if the motion continues the same way

E SAVE FOR THE FINAL MEETING

The desk has a sighting, not a forecast. Everything we learn in Unit 1 serves one question: how fast is 2026-XJ moving, and where will it arrive?

Read the physics: Observation

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

A telescope records where an object is at a known time. That is an observation. A future dot is a prediction, even if it follows a clear pattern.

02 HOW IT WORKS

Give each observed dot a time. Compare the gaps between dots. Equal gaps in equal times suggest constant velocity. Draw a dashed dot for the next position. It is still a guess based on a pattern.

03 WHAT TO CHECK

If a new dot misses the predicted position, check the measurement and the motion model.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart is at 2 m, 3 m, then 4 m, one second apart. Predict its position one second later. 5 m if the motion continues the same way.

05 CONNECT TO XJ

The desk has a sighting, not a forecast. Everything we learn in Unit 1 serves one question: how fast is 2026-XJ moving, and where will it arrive?

SAY IT BEFORE YOU WRITE

Which dot did someone measure? The measured dot is ___. The predicted dot is ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Observation

ES: observación / PT: observação

Meaning in my words	How can you tell?
How would you explain observation to a friend? In my words, this means ___.	The timer reads 2 s as the cart passes the 4 m mark. What did you actually see? I can tell because ___.
Example or sketch	**Non-example and why**
What two labels would you put on a sketch of that moment? My example shows ___.	Someone writes “It will reach 8 m.” Has the camera seen that yet? I would change ___ because ___.

Model

ES: modelo / PT: modelo

Meaning in my words	How can you tell?
How would you explain model to a friend? In my words, this means ___.	A cart moves 2 m each second. What are we assuming if we extend that pattern? I can tell because ___.
Example or sketch	**Non-example and why**
Where would your model put the cart one second after it passes 6 m going right? My example shows ___.	The cart hits a cushion. Would you still extend the same pattern? Why? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Prediction

ES: predicción / PT: previsão

Meaning in my words	How can you tell?
How would you explain prediction to a friend? In my words, this means ___.	Which part of a forecast could we check with the next camera image? I can tell because ___.
Example or sketch	**Non-example and why**
A tracker reads 0, 2, then 4 km a minute apart. Where would you look a minute later? My example shows ___.	Someone calls yesterday's recorded position a prediction. What would you call it? I would change ___ because ___.

Testable question

ES: pregunta comprobable / PT: pergunta testável

Meaning in my words	How can you tell?
How would you explain testable question to a friend? In my words, this means ___.	What would you measure to answer "Does this cart speed up on the ramp?" I can tell because ___.
Example or sketch	**Non-example and why**
What question could two timed pictures of the same cart help you answer? My example shows ___.	"Which cart looks coolest?" What would you change so the class can test it? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw three observed positions and one predicted position. Build a key that separates the two.

Labels you can use: measured · predicted.

Choose zero, direction, and scale.



Tell a partner

The measured dot is ___. The predicted dot is ___.

One more exposure

01

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Separate a sighting from a prediction.

CHOOSE ONE JOB TOGETHER

TELESCOPE SCHEDULER ☐

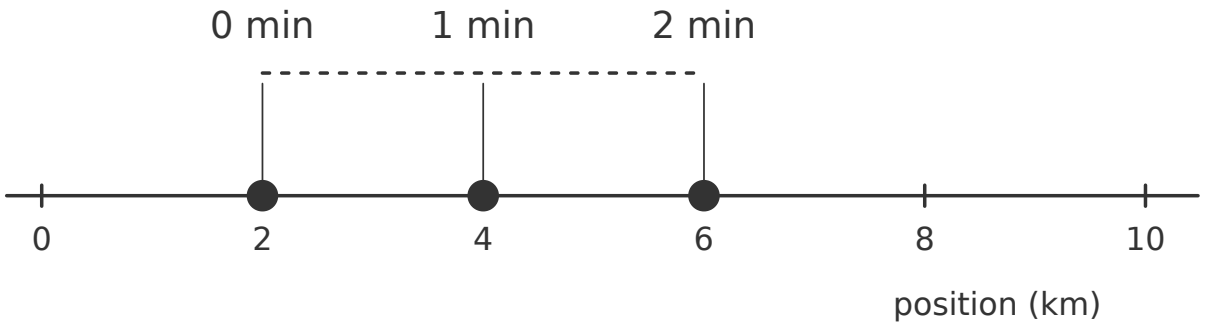
Mark where to aim at minute **3**. Tell the operator which dots were measured and which dot you predicted.

SCIENCE BROADCASTER ☐

Give viewers a two-sentence update. Point to a measured dot, then say where you expect the next one.

Word help: sighting / prediction / pattern

Three telescope fixes are at **2**, **4** and **6 km** at minutes **0**, **1** and **2**. The night crew has one more exposure.



01 OUR FIRST CALL

Where should the camera look next? Use the timed dots in your call.

Aim at ___ because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new sighting fit your prediction? Tell the next crew what to check.

Now we would say ___ because ___.

Switch speakers. Which dot did we measure, and which did we predict?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks